MATH4515 Statistical and Computational Method in Finance

Lecture Note 3

Multivariate analysis, linear regression and principal component analysis

Objective

In topic 1, we focus on modelling the price dynamics of a *single stock* by fitting a probability distribution on the stock return and estimate the parameters using suitable method (say maximum likelihood method).

In working on the investment problem, we need to investigate the price dynamics of *multiple* stocks. However, we cannot apply the method used earlier to model the stock return of the stocks separately since the stocks, especially if they are company stocks in same industry, may be correlated.

The first goal of this chapter is to extend the earlier approach to obtain the *joint distribution* of returns of the multiple stocks.

- Basic concepts about vector of random variables
- Joint distribution
- Expected value and covariance matrix
- Multivariate normal distribution and multivariate t-distribution
- Monte-Carlo simulation for multiple random variables.

Another approach in modeling stock return: Linear regression

Since the returns of the stock are often affected by various market/economic factors and firm-specific risk, one would expect that the stock return can be modelled via factor models, i.e.

$$r_i = g(f_1, f_2, \dots, f_p) + \varepsilon_i,$$

where $f_1, f_2, \dots, f_p$ are the factors chosen, ε_i is the random noise and $g(\cdot)$ is a given function.

To simplify the analysis, we would consider the case when $g(\cdot)$ is a linear function. That is,

$$r_i = \beta_0 + \beta_1 f_1 + \cdots + \beta_p f_p + \varepsilon_i,$$

where $\beta_i s$ are some parameters. This model is known as linear model and it has a wide application in financial engineering.

- **Capital asset pricing model** – The model states that the stock return under market equilibrium can be expressed as

$$r_i = r_f + \beta_{iM}(r_M - r_f) + \varepsilon_i,$$

where r_f is the riskfree interest rate, r_M is the return of market portfolio (collection of all stocks available in the market) and β_{iM} is called beta of the stock.

- **Fama-French three-factor model** – The stock return follows

$$r_i = r_f + \beta_1(r_M - r_f) + \beta_2(r_S - r_L) + \beta_3(r_H - r_L) + \varepsilon_i,$$

where $r_S - r_L$ denotes the difference between the return on a portfolio S of small stock and the return on portfolio L of large stock. $r_H - r_L$ denotes between the return on a portfolio H with high book-to-market stocks and the return on a portfolio L with low book-to-market stocks.

In this Chapter, we shall examine the construction of the linear model using the sample data given. We will also perform statistical analysis on the model constructed (accuracy of the model/ prediction). Finally, we will also examine the construction of linear model via. principal component analysis (PCA).

Quick review on Basic probability theories on vector of random variables

Probability on vectors of random variables: Joint distribution function

We let $\vec{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}$ be a vector of random variables. The joint distribution function of $(X_1, \dots, X_n)$, denoted by $f_X(x_1, \dots, x_n)$ can be defined as follows:

- If $X_i s$ are discrete random variables, the joint distribution function can be expressed as *probability mass function*. That is,
$$f_{\vec{X}}(x_1, x_2 \dots, x_n) = P(X_1 = x_1, X_2 = x_2 \dots, X_n = x_n)$$
- If $X_i s$ are continuous random variables, the joint distribution function quoted as *probability density function* $f_{\vec{X}}(x_1, x_2, \dots, x_n)$ such that
$$P\big(X_1 \in (a_1, b_1), X_2 \in (a_2, b_2), \dots, X_n \in (a_n, b_n)\big)$$
$$= \int_{a_n}^{b_n} \int_{a_{n-1}}^{b_{n-1}} \dots \int_{a_1}^{b_1} f_{\vec{X}}(x_1, \dots, x_n) dx_1 \dots dx_n.$$
for any real numbers a_i, b_i where $a_i < b_i$.

Example 1

The following table shows the joint distribution function of two discrete random variables X and Y.

	$Y=0$	$Y=1$	$Y=2$
$X=0$	$\frac{1}{12}$	$\frac{5}{24}$	$\frac{1}{24}$
$X=1$	$\frac{1}{8}$	$\frac{1}{12}$	$\frac{1}{8}$
$X=2$	$\frac{1}{24}$	$\frac{5}{24}$	$\frac{1}{12}$

(a) Find the probability that $P(X+Y\leq 1)$.

(b) Find the probability that $P(X=1)$.

☺Solution

(a) $P(X+Y\leq 1)=P(X=0,Y=0)+P(X=0,Y=1)+P(X=1,Y=0)$

$=\frac{1}{12}+\frac{5}{24}+\frac{1}{8}=\frac{5}{12}.$

(b) $P(X=1)=P(X=1,Y=0)+P(X=1,Y=1)+P(X=1,Y=2)$

$=\frac{1}{8}+\frac{1}{12}+\frac{1}{8}=\frac{1}{3}.$

Example 2

Suppose that the joint density function of continuous random variables (X, Y) is given by

$$f(x,y) = e^{-(x+y)}, \qquad x \geq 0 \ \ and \ y \geq 0.$$

(a) Calculate the probability that $P(1 \leq X \leq 2)$.

(b) Estimate the probability that $P(X \leq Y \leq 3)$.

(c) Derive the probability density function of $Z = X + Y$

☺ Solution

(a) Using the definition of probability density function, we get

$$P(1 \leq X \leq 2) = P(1 \leq X \leq 2, 0 \leq Y \leq \infty). = \int_1^2 \int_0^\infty e^{-(x+y)} \, dy \, dx$$

$$= \int_1^2 \left[-e^{-(x+y)}\right]_{y=0}^{y=\infty} dx = \int_1^2 e^{-x} dx$$

$$= e^{-1} - e^{-2} \approx 0.232544.$$

(b) The required probability can be computed as

$$P(X \le Y \le 3) = \int_0^3 \left[\int_0^y e^{-(x+y)}\, dx \right] dy = \int_0^3 \left[-e^{-(x+y)} \right]_{x=0}^{x=y} dy$$

$$= \int_0^3 (e^{-y} - e^{-2y}) dy = \left[-e^{-y} + \frac{1}{2} e^{-2y} \right]_0^3 \approx 0.451452.$$

(c) For any $z \ge 0$, we consider

$$F_Z(z) = P(X + Y \le z) = \int_0^z \left[\int_0^{z-x} e^{-(x+y)}\, dy \right] dx$$

$$= \int_0^z \left[-e^{-(x+y)} \right]_{y=0}^{y=z-x} dx = \int_0^z (e^{-x} - e^{-z}) dx$$

$$= \left[-e^{-x} - xe^{-z} \right]_0^z = 1 - e^{-z} - ze^{-z}.$$

Thus the probability density function of $Z = X + Y$ is

$$f_Z(z) = F_Z'(z) = e^{-z} - e^{-z} + ze^{-z} = ze^{-z}.$$

Marginal distribution

Sometimes we may be interested in studying the distribution of a particular random variable X_i. Given the joint distribution function $f_X(x_1, \dots, x_n)$, one can obtain the distribution of X_i (denoted by $f_{X_i}(x_i)$ as follows:

- If X_i is discrete random variable, then we have

$$f_{X_i}(x_i) = P(X_i = x_i) = P\big(X_i = x_i \ \ and \ \ X_j \in (-\infty, \infty) \ \ for \ \ all \ j \neq i\big)$$
$$= \sum_{all\ x_1} \cdots \sum_{all\ x_{i-1}} \sum_{all\ x_{i+1}} \dots \sum_{all\ x_n} f_{\vec{X}}(x_1, x_2, \dots, x_n).$$

- If X_i is continuous random variable, we consider

$$F_{X_i}(x) = P(X_i \leq x) = P\big(X_i = x \ \ and \ \ X_j \in (-\infty, \infty) \ \ for \ \ all \ j \neq i\big)$$
$$= \int_{-\infty}^{x} \left(\int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f_{\vec{X}}(x_1, x_2, \dots, x_n) dx_1 \dots dx_{i-1} dx_{i+1} dx_n \right) dx_i$$

Then the probability density function of X_i can be computed as

$$f_{X_i}(x) = F'_{X_i}(x) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f_{\vec{X}}(x_1, \dots, x_{i-1}, x, x_{i+1} \dots, x_n) dx_1 \dots dx_{i-1} dx_{i+1} dx_n$$

The function $f_{X_i}(x)$ is known as *marginal distribution* of X_i.

Independence of random variables

In general, we say the random variables $X_1, X_2, \dots, X_n$ are independent if and only if

$$P(X_1 \in A_1, X_2 \in A_2, \dots, X_n \in A_n) = \prod_{i=1}^{n} P(X_i \in A_i),$$

Here, A_i is subset of real number.

In terms of joint density function, the independence implies that the joint density function $f_{\vec{X}}(x_1, \dots, x_n)$ can be expressed as

$$f_{\vec{X}}(x_1, \dots, x_n) = f_{X_1}(x_1) f_{X_2}(x_2) \dots, f_{X_n}(x_n),$$

where $f_{X_i}(x_i)$ denotes the probability density function (marginal distribution) of X_i.

Example 3

We consider the random variables X and Y considered in Example 2 which the joint density function is $f(x, y) = e^{-(x+y)}$ for $x, y \geq 0$, we note that

- The marginal distribution of X is $f_X(x) = \int_0^\infty e^{-(x+y)} dy = \left[-e^{-(x+y)}\right]_0^\infty = e^{-x}$. Similarly, the marginal distribution of Y is $f_Y(y) = e^{-y}$.
- One can easily verify that $f(x, y) = e^{-(x+y)} = e^{-x} e^{-y} = f_X(x) f_Y(y)$. This proves that X and Y are independent.

Example 4

The joint density function of X and Y is given by

$$f(x,y) = \begin{cases} \lambda^2 e^{-\lambda y} & if \ \ 0 \le x \le y \\ 0 & if \ otherwise \end{cases},$$

where $\lambda > 0$. Determine if X and Y are independent.

😊 Solution

- The marginal distribution of X is

$$f_X(x) = \int_0^\infty f(x,y)dy = \int_x^\infty \lambda^2 e^{-\lambda y} dy = \left[-\lambda e^{-\lambda y}\right]_x^\infty = \lambda e^{-\lambda x}.$$

- The marginal distribution of Y is

$$f_Y(y) = \int_0^\infty f(x,y)dx = \int_0^y \lambda^2 e^{-\lambda y} dx = \lambda^2 y e^{-\lambda y}.$$

By taking $x = 2$ and $y = 1$, we have

$$f(x,y) = 0 \neq \left(\lambda e^{-2\lambda}\right)\left(\lambda^2 e^{-\lambda}\right) = f_X(x) f_Y(y).$$

So X and Y are not independent.

Change of variable formula

Given a random variable $\vec{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}$, we define $\vec{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix} = \begin{pmatrix} g_1(X_1, \dots, X_n) \\ \vdots \\ g_n(X_1, \dots, X_n) \end{pmatrix}$, where $g_1, g_2, \dots, g_n$ are some functions.

We would like to derive the joint density function of this random variable $\vec{Y}$.

To start with, we consider the one-dimensional case when $n = 1$:

We let $Y = g(X)$. Suppose that the function $g(x)$ is continuously differentiable and strictly monotone (i.e. $g'(x) \neq 0$). Then $g(x)$ is either strictly increasing or strictly decreasing.

- If $g(x)$ is strictly increasing, then the c.d.f of Y is

$$F_Y(y) = P(Y = g(X) \leq y) = P\big(X \leq g^{-1}(y)\big) = \int_{-\infty}^{g^{-1}(y)} f_X(x)dx = F_X\big(g^{-1}(y)\big).$$

  By differentiating both side with respect to y, we get

$$f_Y(y) = F_Y'(y) = F_X'\big(g^{-1}(y)\big)\frac{d}{dy}g^{-1}(y) = f_X\big(g^{-1}(y)\big)\frac{1}{g'\big(g^{-1}(y)\big)}.$$

- If $g(x)$ is strictly decreasing, then the c.d.f of Y is

$$F_Y(y) = P(Y = g(X) \le y) = P\big(X \ge g^{-1}(y)\big) = 1 - F_X\big(g^{-1}(y)\big).$$

By differentiating both side with respect to y, we get

$$f_Y(y) = F_Y'(y) = \cdots = -f_X\big(g^{-1}(y)\big)\frac{1}{g'\big(g^{-1}(y)\big)} \overset{g'(y)<0}{=\!=} f_X\big(g^{-1}(y)\big)\frac{1}{\big|g'\big(g^{-1}(y)\big)\big|}.$$

The above formula can be extended to n-dimensional as follows:

Suppose that the functions $g_1, g_2, \dots, g_n$ are continuously differentiable such that the Jacobian $J = \det\begin{pmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \cdots & \frac{\partial g_1}{\partial x_n} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \cdots & \frac{\partial g_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_n}{\partial x_1} & \frac{\partial g_n}{\partial x_2} & \cdots & \frac{\partial g_n}{\partial x_n} \end{pmatrix} \neq 0$. We let $\vec{h} = \begin{pmatrix} h_1(y_1, \dots, y_n) \\ h_2(y_1, \dots, y_n) \\ \vdots \\ h_n(y_1, \dots, y_n) \end{pmatrix}$ be the inverse function of $\vec{g} = \begin{pmatrix} g_1(x_1, \dots, x_n) \\ \vdots \\ g_n(x_1, \dots, x_n) \end{pmatrix}$ (which exists by inverse function theorem) .

Then the joint density function of $Y_1, Y_2, \dots, Y_n$ is given by

$$f_Y(y_1, y_2, \dots, y_n) = f_X\big(h_1(y_1, \dots, y_n), \dots, h_n(y_1, \dots, y_n)\big)\left(\frac{1}{|J|}\right).$$

Example 5

We let X_1, X_2 be two independent random variables which X_i is exponentially distributed with mean θ_i (i.e. p.d.f. of X_i is $f_{X_i}(x) = \frac{1}{\theta_i} e^{-\frac{x}{\theta_i}}$). Define

$$Y_1 = X_1 + 2X_2 \quad and \;\; Y_2 = X_1 - X_2.$$

Determine the joint density function of $Y_1,\ Y_2$.

😊Solution

The joint density function of X_1, X_2 is

$$f_X(x_1, x_2) = \left(\frac{1}{\theta_1} e^{-\frac{x_1}{\theta_1}}\right)\left(\frac{1}{\theta_2} e^{-\frac{x_2}{\theta_2}}\right) = \frac{1}{\theta_1\theta_2} e^{-\frac{x_1}{\theta_1}-\frac{x_2}{\theta_2}}.$$

We let $g_1(x_1, x_2) = x_1 + 2x_2$ and $g_2(x_1, x_2) = x_1 - x_2$. Note that

- $J = \det\begin{pmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} \end{pmatrix} = \det\begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix} = -3 \neq 0.$

- Note that $X_1 = \underbrace{\frac{1}{3}(Y_1 + 2Y_2)}_{h_1(Y_1,Y_2)}$ and $X_2 = \underbrace{\frac{1}{3}(Y_1 - Y_2)}_{h_2(Y_1,Y_2)}$. So the inverse function of $\vec{g} = (g_1, g_2)$ is $\vec{h} = \big(h_1(y_1, y_2), h_2(y_1, y_2)\big)$

Using change of variable formula, the joint density function of Y_1 and Y_2 is

$$f(y_1, y_2) = f_X\big(h_1(y_1, \dots, y_n), \dots, h_n(y_1, \dots, y_n)\big)\left(\frac{1}{|J|}\right)$$

$$= \frac{1}{3} f_X\left(\frac{1}{3}(y_1 + 2y_2), \frac{1}{3}(y_1 - y_2)\right)$$

$$= \frac{1}{3\theta_1\theta_2} e^{-\frac{y_1+2y_2}{3\theta_1} - \frac{y_1-y_2}{3\theta_2}}$$

$$= \frac{1}{3\theta_1\theta_2} e^{-\left(\frac{1}{3\theta_1}+\frac{1}{3\theta_2}\right)y_1 - \left(\frac{2}{3\theta_1} - \frac{1}{3\theta_2}y_2\right)}.$$

Expected value and covariance matrix

Given a vector of random variable $\vec{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_k \end{pmatrix}$, the expected value of $\vec{X}$ is defined as

$$\mathbb{E}[\vec{X}] = \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_k] \end{pmatrix}.$$

On the other hand, we define the covariance matrix of $\vec{X}$ as

$$Cov(\vec{X}) = \begin{pmatrix} Cov(X_1, X_1) & Cov(X_1, X_2) & \cdots & Cov(X_1, X_k) \\ Cov(X_2, X_1) & Cov(X_2, X_2) & \cdots & Cov(X_2, X_k) \\ \vdots & \vdots & \ddots & \vdots \\ Cov(X_k, X_1) & Cov(X_k, X_2) & \cdots & Cov(X_k, X_k) \end{pmatrix}$$

Besides capturing the variance of each random variable, the covariance matrix also captures the correlations (proxied by covariance) between the random variables.

Using the fact that $Cov(X_i, X_j) = \mathbb{E}\big[(X_i - \mathbb{E}[X_i])(X_j - \mathbb{E}[X_j])\big]$, the covariance matrix can be rewritten as

$$Cov(\vec{X}) = \mathbb{E}\left[\begin{pmatrix} X_1 - \mathbb{E}[X_1] \\ X_2 - \mathbb{E}[X_2] \\ \vdots \\ X_k - \mathbb{E}[X_k] \end{pmatrix} (X_1 - \mathbb{E}[X_1] \quad X_2 - \mathbb{E}[X_2] \quad \cdots \quad X_k - \mathbb{E}[X_k])\right]$$
$$= \mathbb{E}\left[(\vec{X} - \mathbb{E}[\vec{X}])(X - \mathbb{E}[\vec{X}])^T\right].$$

Additional properties of expected value and covariance matrix

We let A be $p \times k$ matrix and $\vec{b} \in \mathbb{R}^p$ be a vector, the

1. $\mathbb{E}\big[A\vec{X} + \vec{b}\big] = A\mathbb{E}\big[\vec{X}\big] + \vec{b}$.
2. $Cov\big(A\vec{X} + \vec{b}\big) = A\, Cov\big(\vec{X}\big)\, A^T$.

Proof of property 2

Note that $A\vec{X} + \vec{b} - \mathbb{E}\big[A\vec{X} + \vec{b}\big] = A\big(\vec{X} - \mathbb{E}[X]\big)$, we deduce that

$$Cov\big(A\vec{X} + \vec{b}\big) = \mathbb{E}\left[A\big(\vec{X} - \mathbb{E}[X]\big)\big(\vec{X} - \mathbb{E}[X]\big)^T A^T\right] = ACov\big(\vec{X}\big)A^T.$$

Multivariate normal distribution

We let X be a normal random variable with mean μ and standard deviation σ. Then the random variable can be expressed as

$$X = \mu + \sigma Z.$$

Using this result, we can define the multivariate normal distribution as follows:

We let $\vec{Z} = \begin{pmatrix} Z_1 \\ \vdots \\ Z_n \end{pmatrix}$ be a vector of independent standard normal random variables (i.e. $Z_i \sim N(0,1)$). We define

$$\vec{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} = \underbrace{\begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{pmatrix}}_{\vec{\mu}} + \underbrace{\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} Z_1 \\ Z_2 \\ \vdots \\ Z_n \end{pmatrix}}_{\vec{Z}} = \begin{pmatrix} \mu_1 + a_{11}Z_1 + \cdots + a_{1n}Z_n \\ \mu_2 + a_{21}Z_1 + \cdots + a_{2n}Z_n \\ \vdots \\ \mu_n + a_{n1}Z_1 + \cdots + a_{nn}Z_n \end{pmatrix}.$$

Then we say $\vec{Y} = \vec{\mu} + A\vec{Z}$ follows multivariate normal distribution with mean $\mathbb{E}[\vec{Y}] = \vec{\mu}$ and covariance matrix $\Sigma = Cov(\vec{Y}) = ACov(\vec{Z})A^T = AA^T$. We write $\vec{Y} \sim N_n(\vec{\mu}, \Sigma)$.

(*Note: Here, Σ is non-negative definite in the sense that $\vec{w}^T \Sigma \vec{w} \geq 0$ for any $\vec{w} \neq 0$.)

Next, we derive the probability density function of $\vec{Y}$. Note that

- Since $Z_i s$ are independent, the joint density function of $\vec{Z}$ is

$$f_{\vec{Z}}(z_1, z_2, \dots, z_n) = \prod_{i=1}^{n} \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{z_i^2}{2}}}_{f_{Z_i}(z_i)} = \frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{\sum_{i=1}^{n} z_i^2}{2}} = \frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{\vec{z}^T \vec{z}}{2}}.$$

- We let $\vec{X} = \vec{\mu} + A\vec{Z} = g(\vec{Z}) = \begin{pmatrix} g_1(\cdot) \\ g_2(\cdot) \\ \vdots \\ g_n(\cdot) \end{pmatrix} = \begin{pmatrix} \mu_1 + a_{11}Z_1 + \cdots + a_{1n}Z_n \\ \mu_2 + a_{21}Z_1 + \cdots + a_{2n}Z_n \\ \vdots \\ \mu_n + a_{n1}Z_1 + \cdots + a_{nn}Z_n \end{pmatrix}$.

  Assuming that the matrix A is invertible, then $\vec{Z} = A^{-1}(\vec{X} - \vec{\mu})$. It follows from change of variable formula that

$$f_X(x_1, x_2, \dots, x_n) = \frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{\left(A^{-1}(\vec{x}-\vec{\mu})\right)^T \left(A^{-1}(\vec{x}-\vec{\mu})\right)}{2}} \left(\frac{1}{|\det A|}\right)$$

$$\overset{\substack{\Sigma = AA^T \\ \det\Sigma = (\det A)^2}}{=} \frac{1}{(2\pi)^{\frac{n}{2}} |\det \Sigma|^{\frac{1}{2}}} e^{-\frac{(\vec{x}-\vec{\mu})^T \Sigma^{-1} (\vec{x}-\vec{\mu})}{2}}.$$

Properties of multivariable normal distribution

In the following, we shall present some properties of the distribution which will be useful for forthcoming statistical analysis. Some of these properties are simply the generalization from the univariate case ($n = 1$).

Property 1

If $\vec{X} \sim N_n(\vec{\mu}, \Sigma)$, then $\vec{a}^T\vec{X} = a_1X_1 + a_2X_2 + \cdots + a_nX_n$ is also normally distributed.

Proof of property 1

Since $\vec{X} = \vec{\mu} + C\vec{Z}$, where $CC^T = \Sigma$, it follows that

$$Y = \vec{a}^T\left(\vec{\mu} + C\vec{Z}\right) = \vec{a}^T\vec{\mu} + \vec{a}^TC\vec{Z}.$$

We write $\vec{a}^TC = (c_1, c_2, \cdots, c_n)$, then $\vec{a}^TC\vec{Z} = c_1Z_1 + c_2Z_2 + \cdots + c_nZ_n$ has normal distribution with mean 0 and variance $\sigma^2 = c_1^2 + c_2^2 + \cdots + c_n^2 = (\vec{a}^TC)(\vec{a}^TC)^T = \vec{a}^TCC^T\vec{a} = \vec{a}^T\Sigma\vec{a}$, i.e, $\vec{a}^TC\vec{Z} \sim N(0, \sigma^2)$.

By writing $\vec{a}^TC\vec{Z} = \sigma Z$ (where $Z \sim N(0,1)$), then we have

$$Y = \vec{a}^T\mu + \sigma Z$$

So Y is normally distributed.

The above result can be extended to the following general case.

Property 2 (Extended version of property 1)

We let $\vec{X} \sim N_n(\vec{\mu}, \Sigma)$. We let A be a $m \times n$ matrix and $\vec{b} \in \mathbb{R}^m$ be a vector. Then $\vec{Y} = A\vec{X} + \vec{b}$ has multivariate normal distribution with $\vec{Y} \sim N_m\left(A\vec{\mu} + \vec{b}, A\Sigma A^T\right)$.

Proof of property 2

Here, we present the proof for the special case when $m = n$. The general case when $m \neq n$ can be proved using moment generating function/characteristic function. I omit the details here.

Since $\vec{X} \sim N_n(\mu, \Sigma)$, then $\vec{X}$ can be written as $\vec{X} = \mu + C\vec{Z}$, where $\vec{Z} = \begin{pmatrix} Z_1 \\ \vdots \\ Z_n \end{pmatrix}$.

$$A\vec{X} + \vec{b} = A\left(\vec{\mu} + C\vec{Z}\right) + \vec{b} = \underbrace{\left(A\vec{\mu} + \vec{b}\right)}_{\mu'} + \underbrace{AC}_{C'}\vec{Z}.$$

As $A\vec{X} + \vec{b} \in \mathbb{R}^n$ under $m = n$, it follows from the definition that $A\vec{X} + \vec{b}$ has multivariate normal distribution.

Property 3

If $\vec{X} \sim N_n(\vec{\mu}, \Sigma)$ and Σ is invertible, then random variable $\vec{Y} = \left(\vec{X} - \vec{\mu}\right)^T \Sigma^{-1} \left(\vec{X} - \vec{\mu}\right)$ has chi-squared distribution with degree of freedom n.

Proof of property 3:

Write $\vec{X} = \vec{\mu} + A\vec{Z}$, where $\vec{Z} = \begin{pmatrix} Z_1 \\ \vdots \\ Z_n \end{pmatrix}$ is a vector of independent standard normal random variables and $AA^T = \Sigma$. Then

$$\vec{Y} = \left(\vec{\mu} + A\vec{Z} - \vec{\mu}\right)^T \Sigma^{-1} \left(\vec{\mu} + A\vec{Z} - \vec{\mu}\right) = \vec{Z}^T A^T (AA^T)^{-1} \vec{A}\vec{Z} = \vec{Z}^T \vec{Z} = \underbrace{Z_1^2 + \cdots + Z_n^2}_{\chi_n^2}.$$

Property 4 (Correlation and independence)

We let $\vec{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_m \end{pmatrix}$ and $\vec{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}$ be two vectors of random variables such that $\overrightarrow{W} = \begin{pmatrix} \vec{X} \\ \vec{Y} \end{pmatrix}$ follows multivariate normal distribution with $\overrightarrow{W} \sim N_{m+n}\left(\underbrace{\begin{pmatrix} \vec{\mu}_X \\ \vec{\mu}_Y \end{pmatrix}}_{\vec{\mu}}, \underbrace{\begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{pmatrix}}_{\Sigma} \right)$.

Then $\vec{X}$ and $\vec{Y}$ are independent if and only if Σ_{XY} and Σ_{YX} are zero matrix.

Remark: Note that Σ_{XY} is a collection of $cov(X_i, Y_j)$ for $i = 1,2,..,m$ and $j = 1,2,\dots,n$. This property states that two normal random variables from the above multivariate normal distribution are independent if the covariance of them is 0.)

Proof of property 4

("⇒" part) If $\vec{X}, \vec{Y}$ are independent, it follows that X_i and Y_j are independent for every $i = 1,2,\dots,m$ and $j = 1,2,\dots,n$. Then we have $cov(X_i, Y_j) = 0$ and Σ_{XY}, Σ_{YX} are zero matrix.

("⇐" part) Given that Σ_{XY} and Σ_{YX} are zero matrix, then the covariance matrix and its inverse are given by

$$\Sigma = \begin{pmatrix} \Sigma_{XX} & 0 \\ 0 & \Sigma_{YY} \end{pmatrix} \Rightarrow \Sigma^{-1} = \begin{pmatrix} \Sigma_{XX}^{-1} & 0 \\ 0 & \Sigma_{YY}^{-1} \end{pmatrix}.$$

Then the joint density function of $\vec{W}$ can be expressed as

$$f_W(x_1, \dots, x_m, y_1, \dots, y_n) = \frac{1}{(2\pi)^{\frac{m+n}{2}} (\det \Sigma)^{\frac{1}{2}}} e^{-\frac{(\vec{w}-\vec{\mu})^T \Sigma^{-1} (\vec{w}-\vec{\mu})}{2}}, \qquad where\ \vec{w} = \begin{pmatrix} \vec{x} \\ \vec{y} \end{pmatrix}$$

$$= \frac{1}{(2\pi)^{\frac{m+n}{2}}(\det\Sigma_{XX}\det\Sigma_{YY})^{\frac{1}{2}}} e^{-\frac{(\vec{x}-\vec{\mu}_X)^T\Sigma_{XX}^{-1}(\vec{x}-\vec{\mu}_X)}{2}-\frac{(\vec{y}-\vec{\mu}_y)^T\Sigma_{YY}^{-1}(\vec{y}-\vec{\mu}_y)}{2}}$$

$$= \underbrace{\left(\frac{1}{(2\pi)^{\frac{m}{2}}(\det\Sigma_{XX})^{\frac{1}{2}}} e^{-\frac{(\vec{x}-\vec{\mu}_X)^T\Sigma_{XX}^{-1}(\vec{x}-\vec{\mu}_X)}{2}}\right)}_{f_X(x_1,x_2,\ldots,x_m)} \underbrace{\left(\frac{1}{(2\pi)^{\frac{n}{2}}(\det\Sigma_{YY})^{\frac{1}{2}}} e^{-\frac{(\vec{y}-\vec{\mu}_y)^T\Sigma_{YY}^{-1}(\vec{y}-\vec{\mu}_y)}{2}}\right)}_{f_Y(y_1,y_2,\ldots,y_n)}$$

It follows that $\vec{X}$ and $\vec{Y}$ are independent by definition.

Remarks of Property 4

This result requires that the random variables are normally distributed. It is not the case if the random variables are not normally distributed.

For example, we let X is uniformly distributed over $[-1,1]$ and define $Y = X^2$. One can show that $cov(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0$ and

$$P\left(X \geq \frac{1}{2} \ \ and \ \ Y = X^2 \geq \frac{1}{4}\right) = P\left(X \geq \frac{1}{2}\right) \neq P\left(X \geq \frac{1}{2}\right) P\left(Y \geq \frac{1}{4}\right).)$$

Property 5 (Joint distribution of independent multivariate normal distribution)

We let $\vec{X}_1$ and $\vec{X}_2$ be two independent random variables such that $\vec{X}_1 \sim N_p(\vec{\mu}_1 . \Sigma_{11})$ and $\vec{X}_2 \sim N_q(\vec{\mu}_2, \Sigma_{22})$. Then $\vec{X} = \begin{pmatrix} \vec{X}_1 \\ \vec{X}_2 \end{pmatrix}$ follows multivariate normal distribution with mean $\vec{\mu} = \begin{pmatrix} \vec{\mu}_1 \\ \vec{\mu}_2 \end{pmatrix}$ and covariance matrix $\begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$.

Proof of property 5

Note that the joint density functions of $\vec{X}_1$ and $\vec{X}_2$ are given by

$$f_{\vec{X}_1} = \frac{1}{(2\pi)^{\frac{p}{2}} (\det \Sigma_{11})^{\frac{1}{2}}} e^{-\frac{(\vec{x}_1 - \vec{\mu}_1)^T \Sigma_{11}^{-1} (\vec{x}_1 - \vec{\mu}_1)}{2}}, \quad f_{\vec{X}_2} = \frac{1}{(2\pi)^{\frac{q}{2}} (\det \Sigma_{22})^{\frac{1}{2}}} e^{-\frac{(\vec{x}_2 - \vec{\mu}_2)^T \Sigma_{22}^{-1} (\vec{x}_2 - \vec{\mu}_2)}{2}}$$

Since $\vec{X}_1$ and $\vec{X}_2$ are independent, it follows from definition that

$$f_X \left(\underbrace{x_{11}, \dots, x_{1p}}_{\vec{x}_1}, \underbrace{x_{21}, \dots, x_{2q}}_{\vec{x}_2} \right) = f_{X_1}(\vec{x}_1) f_{\vec{X}_2}(\vec{x}_2)$$

$$= \left(\frac{1}{(2\pi)^{\frac{p}{2}} (\det \Sigma_{11})^{\frac{1}{2}}} e^{-\frac{(\vec{x}_1 - \vec{\mu}_1)^T \Sigma_{11}^{-1} (\vec{x}_1 - \vec{\mu}_1)}{2}} \right) \left(\frac{1}{(2\pi)^{\frac{q}{2}} (\det \Sigma_{22})^{\frac{1}{2}}} e^{-\frac{(\vec{x}_2 - \vec{\mu}_2)^T \Sigma_{22}^{-1} (\vec{x}_2 - \vec{\mu}_2)}{2}} \right)$$

$$= \frac{1}{(2\pi)^{\frac{p+q}{2}} (\det \Sigma)^{\frac{1}{2}}} e^{-\frac{(\vec{x} - \vec{\mu})^T \sum^{-1} (\vec{x} - \vec{\mu})}{2}}.$$

(*The last equality can be derived by reversing the calculation adopted in the proof of property 4).

One important consequence of this property is related to sum of independent multivariate normal random variables.

Property 6 (Sum of independent normal random variables)

We let $\vec{X}_1, \vec{X}_2 \in \mathbb{R}^p$ be two independent random variables which each $\vec{X}_i$ has multivariate normal with $\vec{X}_i \sim N_p(\vec{\mu}_i, \Sigma_{ii})$. Then

$$\vec{Y} = c_1 \vec{X}_1 + c_2 \vec{X}_2$$

has multivariate normal distribution with $\vec{Y} \sim N_p(c_1 \vec{\mu}_1 + c_2 \vec{\mu}_2, c_1^2 \Sigma_{11} + c_2^2 \Sigma_{22})$.

Proof of property 6

Since $\vec{X}_1$ and $\vec{X}_2$ are independent, it follows that $\vec{X} = \begin{pmatrix} \vec{X}_1 \\ \vec{X}_2 \end{pmatrix}$ has multivariate normal distribution with $\vec{X} \sim N_{2p}\left(\begin{pmatrix} \vec{\mu}_1 \\ \vec{\mu}_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22} \end{pmatrix}\right)$.

Note that $\vec{Y} = c_1\vec{X}_1 + c_2\vec{X}_2$ can be expressed as

$$\vec{Y} = \underbrace{\begin{pmatrix} c_1 & 0 & \cdots & 0 & c_2 & 0 & \cdots & 0 \\ 0 & c_1 & \cdots & 0 & 0 & c_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c_1 & 0 & 0 & \cdots & c_2 \end{pmatrix}}_{C} \begin{pmatrix} X_{11} \\ \vdots \\ X_{1p} \\ X_{21} \\ \vdots \\ X_{2p} \end{pmatrix}.$$

Using the property 1, we deduce that

$$c_1\vec{X}_1 + c_2\vec{X}_2 = \vec{Y} \sim N_p\left(C\begin{pmatrix} \vec{\mu}_1 \\ \vec{\mu}_2 \end{pmatrix}, C\begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22} \end{pmatrix}C^T\right)$$

$$\Rightarrow c_1\vec{X}_1 + c_2\vec{X}_2 \sim N_p(c_1\vec{\mu}_1 + c_2\vec{\mu}_2, c_1^2\Sigma_{11} + c_2^2\Sigma_{22})$$

Quick applications of multivariate normal distribution

Example 6 (Portfolio management)

There are 3 risky assets available in the market. You are given that

- The monthly return of the risky assets has multivariate normal distribution.
- The expected return and covariance matrix of the risky assets' returns are

$$\mu = \begin{pmatrix} 0.03 \\ 0.05 \\ 0.08 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 0.01 & 0.01 & 0 \\ 0.01 & 0.04 & -0.02 \\ 0 & -0.02 & 0.09 \end{pmatrix}.$$

The investor wants to invest his capital into these three risky assets and aims to achieve a minimum monthly return of 4%.

Suppose that the investor chooses to invest 30% of capital in asset 1, 50% of capital in asset 2 and 20% of capital in asset 3, estimate the probability that his/her investment goal can be achieved.

☺Solution

We let w_1, w_2, w_3 be the proportion of capital invested in asset 1, asset 2 and asset 3 respectively (and $w_1 + w_2 + w_3 = 1$).

Then the monthly return of the portfolio can be expressed as

$$r_P = w_1 r_1 + w_2 r_2 + w_3 r_3.$$

Since $\vec{r} = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix} \sim N(\vec{\mu}, \Sigma)$, it follows that $r_P = \begin{pmatrix} w_1 & w_2 & w_3 \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix} = \vec{w}^T \vec{r}$ has normal distribution with $r_P \sim N(\vec{w}^T \vec{\mu},\ \vec{w}^T \Sigma \vec{w})$.

Using the given parameter $\vec{\mu}$ and $\vec{w} = \begin{pmatrix} 0.3 \\ 0.5 \\ 0.2 \end{pmatrix}$, we have $r_P \sim N(0.05, 0.0135)$.

Therefore, the required probability can be computed as

$$P(r_p \geq 0.04) = P\left(Z \geq \frac{0.04 - 0.05}{\sqrt{0.0135}} \approx -0.08607\right) \approx 0.534293.$$

Example 7 (Checking the independence)

A random variable $\vec{X} = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}$ has multivariate normal distribution with mean $\vec{\mu} = \begin{pmatrix} -3 \\ 1 \\ 4 \end{pmatrix}$ and covariance matrix $\Sigma = \begin{pmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

(a) Determine if X_2 and X_3 are independent

(b) Determine if $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_3$ are independent.

😊 Solution

(a) Since $\sigma_{23} = cov(X_2, X_3) = 0$, so X_2, X_3 are independent.

(b) Note that $\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}}_{A} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}$, it follows that (Y_1, Y_2) follows multivariate normal with covariance matrix $A\Sigma A^T = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$. Since $cov(Y_1, Y_2) = -1 \neq 0$, so Y_1, Y_2 are not independent.

Example 8

We let X be a population with mean $\mathbb{E}[X] = \mu$ and variance $Var(X) = \sigma^2$. A random sample $\{X_1, X_2, \dots, X_n\}$ is drawn from the population.

Recall that we can estimate μ and σ^2 by sample mean and sample variance:

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Suppose that X is normally distributed, prove that $\bar{X}$ and s^2 are independent.

□Solution

One can verify that $\sum_{i=1}^{n}(X_i - \bar{X}) = n\bar{X} - n\bar{X} = 0$, it follows that

$$X_n - \bar{X} = -(X_1 - \bar{X}) - (X_2 - \bar{X}) - \cdots - (X_{n-1} - \bar{X}).$$

So the sample variance can be expressed as

$$s^2 = \sum_{i=1}^{n-1}(X_i - \bar{X})^2 + \left(\sum_{i=1}^{n-1}(X_i - \bar{X})\right)^2.$$

To prove $\bar{X}$ and s^2 are independent, it suffices to prove the random vectors

$$\bar{X} \quad and \quad \vec{Y} = \begin{pmatrix} X_1 - \bar{X} \\ \vdots \\ X_n - \bar{X} \end{pmatrix}$$

are independent.

We note that

$$\begin{pmatrix} \bar{X} \\ \vec{Y} \end{pmatrix} = \begin{pmatrix} \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} & \frac{1}{n} \\ 1-\frac{1}{n} & -\frac{1}{n} & \cdots & -\frac{1}{n} & -\frac{1}{n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -\frac{1}{n} & -\frac{1}{n} & \cdots & 1-\frac{1}{n} & -\frac{1}{n} \end{pmatrix} \underbrace{\begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_{n-1} \\ X_n \end{pmatrix}}_{\vec{X}}.$$

Since $X_1, X_2, .., X_n$ are independent normal distribution, so $\vec{X}$ has multivariate normal distribution and hence $\begin{pmatrix} \bar{X} \\ \vec{Y} \end{pmatrix}$ has multivariate normal distribution as well.

We let $A = \begin{pmatrix} \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} \end{pmatrix}$ and $B = \begin{pmatrix} 1-\frac{1}{n} & -\frac{1}{n} & \cdots & -\frac{1}{n} & -\frac{1}{n} \\ -\frac{1}{n} & 1-\frac{1}{n} & \cdots & -\frac{1}{n} & -\frac{1}{n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -\frac{1}{n} & -\frac{1}{n} & \cdots & 1-\frac{1}{n} & -\frac{1}{n} \end{pmatrix}$, we have $\bar{X} = A\vec{X}$ and $\vec{Y} = B\vec{X}$. Using the property of covariance, we have

$$\Sigma_{\bar{X}\vec{Y}} = Cov(\bar{X}, \vec{Y}) = Cov(A\vec{X}, B\vec{X}) = \mathbb{E}\left[(A\vec{X} - \mathbb{E}[A\vec{X}])(B\vec{X} - \mathbb{E}[B\vec{X}])^T\right]$$

$$= A\underbrace{\mathbb{E}\left[(\vec{X} - \mathbb{E}[\vec{X}])(\vec{X} - \mathbb{E}[\vec{X}])^T\right]}_{Cov(\vec{X})}B^T = A\begin{pmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{pmatrix}B^T$$

$$= \sigma^2\begin{pmatrix} \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} \end{pmatrix}\begin{pmatrix} 1-\frac{1}{n} & -\frac{1}{n} & \cdots & -\frac{1}{n} \\ -\frac{1}{n} & 1-\frac{1}{n} & \cdots & -\frac{1}{n} \\ \vdots & \vdots & \ddots & \vdots \\ -\frac{1}{n} & -\frac{1}{n} & \cdots & 1-\frac{1}{n} \\ -\frac{1}{n} & -\frac{1}{n} & \cdots & -\frac{1}{n} \end{pmatrix} = \begin{pmatrix} 0 & 0 & \cdots & 0 \end{pmatrix}$$

So we conclude from property 4 that $\bar{X}$ and $\vec{Y}$ (or $\bar{X}$ and s^2) are independent.

Some comments about the use of multivariate normal distribution

- Using multivariate normal distribution, one can obtain the multivariate version of *Central limit theorem* as follows:
  We let $\vec{X}_1, \vec{X}_2, \dots, \vec{X}_n$ (where $\vec{X}_i \in \mathbb{R}^p$) be n independent and identically distributed random variables with $\mathbb{E}[\vec{X}_i] = \vec{\mu}$ and $Cov(\vec{X}_i) = \Sigma$. Then when $n \to \infty$, $\frac{1}{n}(\vec{X}_1 + \vec{X}_2 + \cdots + \vec{X}_n)$ is approximately multivariate normal with mean $\vec{\mu}$ and covariance matrix $\frac{1}{n}\Sigma$.
- (Model fitting to stock return)
  We let $\vec{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}$ be a multivariate normal random variable. Then it follows from the property of multivariate normal distribution that each X_i is normally distributed.
  This result reveals that multivariate normal distribution cannot be a good fit to stock return since the stock return may not necessary be normally distributed.

Multivariate t-distribution

As we have seen in earlier chapter that normal distribution may not be a good candidate for modelling the distribution of stock return. Instead, it appears that general t-distribution is a more appropriate candidate since it has heavier tail. So we shall explore the "multi-dimensional" version of t-distribution.

Definition of multivariate t-distribution

We let χ_v^2 be the chi-square random variable with degree of freedom v and $\vec{Z} = \begin{pmatrix} Z_1 \\ \vdots \\ Z_n \end{pmatrix}$ be a vector of random variable (independent of χ_v^2) which has *multivariate normal distribution* with $N_n(0, \Lambda)$.

Then we say a random variable $\vec{Y}$ has multi-variate t-distribution if and only if it can be expressed as

$$\vec{Y} = \vec{\mu} + \sqrt{\frac{v}{\chi_v^2}}\vec{Z},$$

where $\vec{\mu} = (\mu_1 \ \mu_2 \ \cdots \ \mu_n)^T$ is a constant vector.

Some remarks about multivariate t-distribution

- The mean and covariance matrix of $\vec{Y}$ can be computed as

$$\mathbb{E}[\vec{Y}] = \vec{\mu}, \qquad Cov(\vec{Y}) = \frac{v}{v-2}\Lambda, \qquad v > 2.$$

- We let $X = \vec{w}^T\vec{Y} = w_1Y_1 + w_2Y_2 + \cdots + w_nY_n$. Then it can be shown that X has general t-distribution since

$$\vec{X} = \vec{w}^T\vec{Y} = \vec{w}^T\left(\vec{\mu} + \sqrt{\frac{v}{\chi_v^2}}\vec{Z}\right) = \vec{w}^T\vec{\mu} + \sqrt{\frac{v}{\chi_v^2}}\vec{w}^T\vec{Z}.$$

Note that $\vec{w}^T\vec{\mu} = w_1\mu_1 + \cdots + w_n\mu_n = \mu^*$ and $\vec{w}^T\vec{Z} \sim N\left(0, \underbrace{\vec{w}^{\mathrm{T}}\Lambda\vec{w}}_{\sigma^2}\right)$, then

$$\vec{X} = \mu^* + \sigma\frac{Z^*}{\underbrace{\sqrt{\chi_v^2/v}}_{T}}, \quad where \;\; Z^* \sim N(0,1).$$

If we take Y_i be the return of ith asset and w_i be the proportion of capital invested in i^{th} asset, then $\vec{X}$ represents the return of the portfolio. If the returns of those n stocks follows multivariate t-distribution, it follows that the portfolio return also follows t-distribution.

Model fitting using multivariate normal distribution and multivariate t-distribution

Similar to the univariate case, one can estimate the parameters in these two multivariate distribution using the sample data and maximum likelihood method. That is, we let $f_{\vec{X}}(x_1, x_2, \dots, x_p|\theta)$ be the joint distribution and $\{\vec{X}_1, \vec{X}_2, \dots, \vec{X}_n\}$ be the sample data, we choose the parameter θ such that the loglikelihood function (given by $l(\theta) = \ln L(\theta) = \ln\left(\prod_{i=1}^{n} f_{\vec{X}}(\vec{X}_i)\right) = \sum_{i=1}^{n} \ln f_{\vec{X}}(\vec{X}_i)$) is maximized.

Determining MLE of multivariate normal distribution using RStudio

Practically, it is not easy to derive the explicit formula for MLE for multivariate distribution since the probability density function of multivariate distribution is quite tedious.

In RStudio, one can obtain the required MLE easily using the following command:

"mvnorm.mle(x)"

Here, "x" denote the sample data (in forms of matrix).

Remark: The package "Rfast" is required in order to use this command.

Example 9

As an example, we consider 5 US stocks (LUV, LLY, BRK-B, ADBE, DAL) and would like to model their log-return by multivariate normal distribution. To do so, we collect historical daily return of these 5 US stocks from 1st Jan, 2011 to 31st Dec, 2019 (see Return_USstock.xlsx).

Here is the command for finding MLE in multivariate normal distribution:

```
library(Rfast)
x<-Return_USstock[c(2,3,4,5,6)] #Extract the data x<-na.omit(x) #Remove "NA" from data
set
x<-as.matrix(x) #Declare the dataframe as matrix
fitresult<-mvnorm.mle(x) #Find MLE
print(fitresult) #Show the output
```

The output is given as follows:

```
$loglik
[1] 32710.7

$mu
[1] 0.0006529132 0.0007177640 0.0004563810 0.0010392357 0.0007299806

$sigma
              LUV          LLY          BRK-B         ADBE          DAL
LUV   3.087916e-04 5.698389e-05 8.883176e-05 1.051937e-04 2.611113e-04
LLY   5.698389e-05 1.651517e-04 5.806781e-05 6.801592e-05 7.136994e-05
BRK-B 8.883176e-05 5.806781e-05 1.224409e-04 9.052820e-05 1.014501e-04
ADBE  1.051937e-04 6.801592e-05 9.052820e-05 2.817201e-04 1.266348e-04
DAL   2.611113e-04 7.136994e-05 1.014501e-04 1.266348e-04 4.569013e-04
```

MLE of $\vec{\mu}$

MLE of Σ

Determining MLEs in multivariate t-distribution

The procedure is more complicated in this case since there is no single command for finding all parameters in multivariate t-distribution.

In RStudio, one can use following command to find MLEs of $\vec{\mu}$ and Λ in t-distribution *given a fixed value of ν (degree of freedom).*

"mvt.mle(x,v)"

Here, "x" denotes the sample data and "v" denotes the degree of freedom.

To find the MLEs of parameters in multivariate t-distribution, one can follow the following procedures:

Step 1: Choose a suitable range of degree of freedom ν (i.e. $a \leq \nu \leq b$).

Step 2: For each value of ν in $[a, b]$, find the MLEs of $\vec{\mu}$ and Λ and calculate the value of the loglikelihood function (included in the output of "mvt.mle(x,v)" as "$loglik")

Step 3: Choose ν with the largest value of loglikelihood function and calculate the corresponding MLEs of μ and Λ.

Example 10 (MLE of multivariate-t distribution)

We consider the three US stocks (LLY, BRK-B and ADBE) in Example 8 and would like to model their log-return using multivariate t-distribution. The following table summarizes the D-value ($\sup_{-\infty<x<\infty} |F_n(x) - F(x)|$), p-value and degree of freedom of the stocks under univariate ks-test.

	D-value	p-value	Degree of freedom
LUV	0.011906	0.9056	4.278
LLY	**0.01257**	**0.8671**	**3.495**
BRK-B	**0.013373**	**0.8133**	**3.779**
ADBE	**0.011081**	**0.944**	**3.548**
DAL	0.015251	0.6687	4.346

**Remark*

We consider the stocks “LLY”, “BRK-B”, “ADBE” only because they have similar level of degree of freedom and may fit well to multivariate t-distribution.

One can obtain the MLEs of the using the following program:

```
x<-Return_USstock[c(3,4,5)] #Store the target data
x<-na.omit(x)               #remove "NA" in the dataset
x<-as.matrix(x)             #ensure that the data is in matrix form
df=seq(3,5,0.001)           #set the range of degree of freedom
n=length(df)
loglik=rep(0,n)
#Used to store the value of loglikelihood function for different level of df

for(i in 1:n){
  fitres=mvt.mle(x,v=df[i])
  loglik[i]=fitres$loglik
}                    #For each df, calculate the MLE and loglikelihood function

loglik<-as.matrix(loglik)
dfmax=df[argmax(loglik,rows=FALSE)] #Find the maximizer
fitMLE=mvt.mle(x,v=dfmax) #Estimate the MLE of the parameters
print(fitMLE)
```

The output is shown below:

```
> print(dfmax)
[1] 3.699
> print(fitMLE)
$iters
[1] 9
$loglik
[1] 20881.87
$location
[1] 0.0008970936 0.0004077215 0.0013483571
$scatter
              [,1]          [,2]          [,3]
[1,] 8.151106e-05 2.901638e-05 3.704314e-05
[2,] 2.901638e-05 6.181469e-05 4.651803e-05
[3,] 3.704314e-05 4.651803e-05 1.352983e-04
```

MLE of ν

MLE of $\vec{\mu}$

MLE of Λ

Monte Carlo simulation on multivariate distribution

In this section, we explore how to compute $\mathbb{E}\left[f\left(\vec{X}\right)\right] = \mathbb{E}[f(X_1, X_2, \dots, X_n)]$ using Monte-Carlo simulation.

Roughly speaking, the procedure is very similar to the case for single random variable:

Step 1: Get a set of random sample of size N (denoted by $\vec{X}^{(1)}, \vec{X}^{(2)}, \dots, \vec{X}^{(n)}$, where $\vec{X}^{(k)} = \left(X_1^{(k)}, X_2^{(k)}, \dots., X_n^{(k)}\right)^T$ from the distribution of $\vec{X}$.

Step 2: For each sample data $\vec{X}^{(k)}$, we estimate $f^{(k)} = f\left(\vec{X}^{(k)}\right)$.

Step 3: Then we estimate $\mathbb{E}\left[f\left(\vec{X}\right)\right]$ through sample mean. That is,

$$f\left(\vec{X}\right) = \frac{f^{(1)} + f^{(2)} + \cdots + f^{(N)}}{N}$$

One technical problem is that the random variables $(X_1, X_2, \dots, X_n)$ are correlated but the random sample $(X_1^{(i)}, \dots, X_n^{(i)})$ generated directly from computer program are often independent (so that they are uncorrelated).

Algorithm for generating correlated random variable (for multivariate normal distribution and multivariate t-distribution only)

Suppose that the random variable $\vec{X}$ follows one of these two distributions, then one can generate the required random sample using the definition of the corresponding distribution.

Case 1: If $\vec{X}$ follows multivariate normal distribution with $\vec{X} \sim N(\vec{\mu}, \Sigma)$

Step 1: We first find the matrix A such that $AA^T = \Sigma$

For simplicity, we consider the case for $n = 2$ so that $\Sigma \overset{\sigma_{12}=\sigma_{21}}{=} \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix}$.

We let $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, then we expend the equation and obtain

$$\begin{pmatrix} a_{11}^2 + a_{12}^2 & a_{11}a_{21} + a_{12}a_{22} \\ a_{11}a_{21} + a_{12}a_{22} & a_{21}^2 + a_{22}^2 \end{pmatrix} = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix}.$$

By comparing the entries of two matrices, we obtain **3** equations for $a_{11}, a_{12}, a_{21}, a_{22}$ and one of the unknowns is a *free parameter*.

Because of this flexibility, we can take $a_{12} = 0$. Together with the fact that $\sigma_{11} = \sigma_1^2$, $\sigma_{22}^2 = \sigma_2^2$ and $\sigma_{12} = \rho_{12}\sigma_1\sigma_2$. So that the equations can be simplified into

$$\begin{cases} a_{11}^2 = \sigma_{11} = \sigma_1^2 \\ a_{11}a_{21} = \sigma_{12} = \rho_{12}\sigma_1\sigma_2 \\ a_{21}^2 + a_{22}^2 = \sigma_{22} = \sigma_2^2 \end{cases} \Rightarrow \begin{cases} a_{11} = \sigma_1 \\ a_{21} = \rho_{12}\sigma_2 \\ a_{22} = \sqrt{1 - \rho_{12}^2\sigma_2} \end{cases}.$$

So we obtain the matrix $A = \begin{pmatrix} a_{11} & 0 \\ a_{21} & a_{22} \end{pmatrix}$. (*One can extend this to the general when there are n correlated random variable).

Step 2: Generate two independent standard normal random variable $Z_1^{(i)}$ and $Z_2^{(i)}$ (which can be done easily using computer software).

Step 3: Compute $\vec{X}^{(i)}$ by

$$\vec{X}^{(i)} = \begin{pmatrix} X_1^{(i)} \\ X_2^{(i)} \end{pmatrix} = \underbrace{\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}}_{\vec{\mu}} + \underbrace{\begin{pmatrix} a_{11} & 0 \\ a_{21} & a_{22} \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix}}_{\vec{Z}}.$$

According to definition of multivariate normal distribution, it follows that $\vec{X}^{(i)}$ can be seen as random sample generated from multivariate normal distribution.

Case 2: If $\vec{X}$ follows multivariate t-distribution with $\vec{X} \sim t_v(\vec{\mu}, \Sigma)$

Suppose that the parameters $\vec{\mu}, v, \Sigma$ are given (can be estimated from sample data using maximum likelihood method), one can generate the random sample using the definition directly.

Step 1: Calculate a random sample $(\chi_v^2)^{(i)}$ from chi-square distribution.

Step 2: Calculate a random sample $\vec{Z}^{(i)} = \begin{pmatrix} Z_1 \\ \vdots \\ Z_n \end{pmatrix}$ from multivariate normal distribution with $N(0, \Sigma)$ using the algorithm as in Case 1.

Step 3: Compute $\vec{X}^{(i)} = \begin{pmatrix} X_1^{(i)} \\ \vdots \\ X_n^{(i)} \end{pmatrix}$ by

$$\vec{X}^{(i)} = \vec{\mu} + \sqrt{\frac{v}{(\chi_v^2)^{(i)}}} \vec{Z}^{(i)}.$$

Then $\vec{X}^{(i)}$ is seen to be random sample generated from multivariate t-distribution.

Example 11

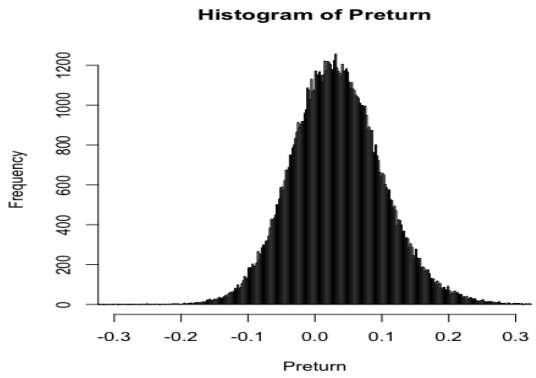
We revisit the Example 10. We consider a portfolio which 20% of capital is invested in LLY, 30% of capital is invested in BRK-B and 50% of capital is invested in ADBE. Assuming that the stock return follows multivariate t-distribution.

Goal: Examine the distribution of portfolio return after 1 months (30 days).

To do so, we first perform Monte Carlo simulation and simulate a sample of 30-days portfolio return:

```
n=100000  #Number of sample data
t=30      #Length of investment periods (in days)
Preturn=rep(0,n) #Portfolio return
for (i in 1:n){
  w=rep(0,3)  #Initial portfolio allocation
  w[1]=0.2
  w[2]=0.3
  w[3]=0.5
  for (j in 1:t){
    x<-rmvt(1,mu=fitMLE$location,sigma=fitMLE$scatter,v=dfmax)
    w=w*(exp(x)) #Calculate the updated value of each components
  }
  Preturn[i]=sum(w)-1;
  print(i)}
```

The histogram of the portfolio return and basic statistics are given as follows:



Basic statistics of 30-days portfolio return

Mean = 3.355%

Median = 3.109%

25th Percentile = -1.240%

75th Percentile = 7.660%

Minimum return = -35.1497%

Maximum return = 124.6166%

On the other hand, we are also interested in finding the probability that the investor will lose more than 3% from this investment. Using the sample data, one can estimate the probability by $\hat{p} = \frac{\mathbf{1}_{\{r_1<-3\%\}}+\cdots+\mathbf{1}_{\{r_n<-3\%\}}}{n}$.

```
> y<-ifelse(Preturn<(-0.03),1,0)
> estimate=mean(y)
> SE=sd(y)/sqrt(n)
> PSE=SE/estimate
```

Using the above algorithm, the probability p is estimated as

$$\hat{p} \approx 17.02\%$$

$$(Standard\ error = 0.1187917\%,\ \ Percentage\ error = 0.6986923\%)$$

Linear regression model

Roughly speaking, linear regression model is a class of model which describe the dependence between a variable y and other factors (proxied by x_i).

The general linear regression model can be presented as follows:

$$y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \cdots + \beta_p x_{tp} + \varepsilon_t,$$

where

- y_t denotes the output variable (formally known as *response variable*)
- $x_{t1}, x_{t2}, \ldots, x_{tp}$ be input variables (called *predictor variables*)
- ε_t is a random error with $\mathbb{E}[\varepsilon_t] = 0$ and $Var(\varepsilon_t) = \sigma^2$.
- $\beta_1, \beta_2, \ldots, \beta_p$ are coefficients of each input variables.

This model is often adopted in studying the problem related to stock return.

Example – Capital Asset Pricing Model (CAPM)

Under this model, the stock return over a period (denoted by r_i) can be expressed as

$$r_i - r_f = \beta\left(r_M - r_f\right) + \varepsilon_i,$$

where r_M and r_f denote the return of market portfolio and riskfree asset respectively.

Determination of coefficients in linear regression model

Given a set of sample data $\{y_i, x_{i1}, x_{i2}, \dots, x_{ip}\}_{1\le i\le n}$, our goal is to determine the β_is such that the following square error (denoted by RSS) is minimized:

$$RSS_p = \sum_{i=1}^{n} \left(\underbrace{y_i}_{\substack{actual\\ value}} - \underbrace{(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip})}_{\substack{predicted\ value\\ (Linear\ regression)}} \right)^2$$

This can be done by considering the first-order condition:

$$\begin{cases} \frac{\partial RSS_p}{\partial \beta_0}|_{\beta_0=\hat{\beta}_0} = 0 \\ \frac{\partial RSS_p}{\partial \beta_1}|_{\beta_1=\hat{\beta}_1} = 0 \\ \vdots \\ \frac{\partial RSS_p}{\partial \beta_p}|_{\beta_p=\hat{\beta}_p} = 0 \end{cases} \Rightarrow \begin{cases} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip}) = 0 \\ \sum_{i=1}^{n} x_{i1}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip}) = 0 \\ \vdots \\ \sum_{i=1}^{n} x_{ip}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip}) = 0 \end{cases}.$$

To express the equations in a more convenient way, we define the following matrices:

$$\hat{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} \in \mathbb{R}^{p+1}, \qquad \vec{Y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^{n}, \qquad X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix} \in \mathbb{R}^{n\times(p+1)}.$$

Then the above system of equations can be expressed as

$$\begin{cases} \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip}\right) = 0 \\ \sum_{i=1}^{n} x_{i1}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip}\right) = 0 \\ \vdots \\ \sum_{i=1}^{n} x_{ip}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_p x_{ip}\right) = 0 \end{cases} \Rightarrow \begin{cases} (1 \quad 1 \quad \cdots \quad 1)\left(\vec{Y} - X\hat{\beta}\right) = 0 \\ (x_{11} \quad x_{21} \quad \cdots \quad x_{n1})\left(\vec{Y} - X\hat{\beta}\right) = 0 \\ \vdots \\ (x_{1p} \quad x_{2p} \quad \cdots \quad x_{np})\left(\vec{Y} - X\hat{\beta}\right) = 0 \end{cases}$$

$$\Rightarrow X^T\left(\vec{Y} - X\hat{\beta}\right) = \vec{0} \Rightarrow \hat{\beta} = (X^T X)^{-1} X^T \vec{Y}.$$

Here, $\hat{\beta}$ is known least square estimate (OLS) of $\vec{\beta}$.

Example 12 (A simple numerical example)

Suppose that we would like to fit a linear regression model

$$y_t = \beta_0 + \beta_1 x_t + \varepsilon_t$$

to the following data

x_t	1	2	3	4	5
y_t	1.442646	3.927477	10.858773	7.294396	12.101381
x_t	6	7	8	9	10
y_t	15.793899	11.518489	20.976623	19.937705	17.144241

Determine the least square estimates for β_0 and β_1.

😊 Solution

Let $X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ \vdots & \vdots \\ 1 & 10 \end{pmatrix}$. We get $X^T X = \begin{pmatrix} 10 & 55 \\ 55 & 385 \end{pmatrix} \Rightarrow (X^T X)^{-1} = \frac{1}{825}\begin{pmatrix} 385 & -55 \\ -55 & 10 \end{pmatrix}$.

Take $\vec{Y} = \begin{pmatrix} 1.442646 \\ 3.927477 \\ \vdots \\ 17.144241 \end{pmatrix}$. Then the least square estimate of $\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}$ is

$$\hat{\beta} = (X^T X)^{-1} X^T \vec{Y} = \begin{pmatrix} 1.421547 \\ 1.941457 \end{pmatrix}.$$

Remark

Given the set of data, one can perform the linear regression in R using the following command

summary(lm(yvariable~xvariable, data=))

The output is shown as below

```
Call:
lm(formula = y ~ x, data = L2_testdata1)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.4216     2.0733   0.686 0.512316
x             1.9415     0.3341   5.810 0.000401 ***
```

Statistical inference on linear regression model

In order to execute the statistical inference on $\hat{\beta}$ or the prediction $\hat{Y}$, we shall derive some statistical properties of the OLS estimates.

We consider the linear regression model of the following form:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i, \qquad 1 \le i \le n, \qquad p+1 \le n$$

or in matrix form,

$$\vec{Y} = X\vec{\beta} + \vec{\varepsilon}.$$

To facilitate the analysis, we will make the following assumptions:

- For $1 \le i \le n$, $\varepsilon_i s$ are independent random variable such that $\varepsilon_i \sim N(0, \sigma^2)$.
  (*Note: In general, it is possible to extend to the case when $\varepsilon_1, \ldots, \varepsilon_n$ follows multivariate normal distribution.)
- x_{ij} are non-random constants and the matrix X has full rank $p+1$.
  (*Note: This implies that $rank(X^T X) = rank(X) = p+1$ and $X^T X$ is invertible)

Under these assumptions, one can establish the following properties

Property 1 (Distribution of OLS estimate $\widehat{\boldsymbol{\beta}}$)

$\hat{\beta} \sim N_{p+1}\left(\vec{\beta}, \sigma^2 (X^T X)^{-1}\right)$. That is, $\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \vdots \\ \hat{\beta}_p \end{pmatrix}$ follows multivariate normal distribution with mean $\vec{\beta}$ and covariance matrix $\sigma^2 (X^T X)^{-1}$.

Property 2 (Estimating σ^2 in random error term)

The unbiased estimator of σ^2 is

$$s^2 = \frac{1}{n-p-1} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

Property 3: The estimates $\hat{\beta}$ and s^2 are independent.

Property 4: (About the statistical property of s^2)

$$\frac{(n-p-1)s^2}{\sigma^2} = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sigma^2} \sim \chi^2_{n-p-1}.$$

Proof of Property 1

Recall that $\vec{Y} = X\vec{\beta} + \vec{\varepsilon} = X\vec{\beta} + \sigma\vec{Z}$, where $\vec{Z} = \begin{pmatrix} Z_1 \\ Z_2 \\ \vdots \\ Z_n \end{pmatrix}$ is a vector of independent standard normal random variable.

Using the formula of the OLS estimate of $\hat{\beta}$, we have

$$\hat{\beta} = (X^T X)^{-1} X^T \vec{Y} = (X^T X)^{-1} X^T \left(X\vec{\beta} + \sigma\vec{Z}\right) = \underbrace{\vec{\beta}}_{\vec{\mu}} + \underbrace{\sigma[(X^T X)^{-1} X^T]}_{A} \vec{Z}.$$

Hence, it follows from the definition that $\hat{\beta}$ has multivariate normal distribution with mean $\vec{\mu}$ and covariance matrix $AA^T = \sigma^2[(X^T X)^{-1} X^T][(X^T X)^{-1} X^T]^T = \cdots = \sigma^2 (X^T X)^{-1}$.

Proof of Property 2

To facilitate the derivation, we define the vector $\vec{e} = \begin{pmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{pmatrix} = \vec{Y} - \hat{Y}$, where $\hat{Y} = X\hat{\beta}$.

We define a matrix $H = X(X^TX)^{-1}X^T$, one can verify that

- $\hat{Y} = X\hat{\beta} = X(X^TX)^{-1}X^T\vec{Y} = H\vec{Y}$;
- $\vec{e} = \vec{Y} - \underbrace{H\vec{Y}}_{\hat{Y}} = (I_n - H)\vec{Y}$, where I_n denotes $n \times n$ identity matrix.
- $H^2 = X(X^TX)^{-1}X^T(X(X^TX)^{-1}X^T) = X(X^TX)^{-1}X^T = H$ and $(I_n - H)^2 = I_n - H$.

We proceed to the proof of the property:

We would like to compute the value of $\mathbb{E}[\sum_{i=1}^{n}(y_i - \hat{y}_i)^2]$. Note that

$$\mathbb{E}\left[\sum_{i=1}^{n}(y_i - \hat{y}_i)^2\right] = \sum_{i=1}^{n}\mathbb{E}[e_i^2] = \sum_{i=1}^{n}Var(e_i) \dots\dots (*)$$

The last inequality follows from the fact that $\mathbb{E}[e_i] = \mathbb{E}[y_i] - \mathbb{E}[\hat{y}_i] = \beta_0 + \beta_1 x_{i1} + \cdots +$

$$\beta_p x_{ip} - \left(\underbrace{\beta_0}_{\mathbb{E}[\widehat{\beta}_0]} + \underbrace{\beta_1}_{\mathbb{E}[\widehat{\beta}_1]} x_{i1} + \cdots + \underbrace{\beta_p}_{\mathbb{E}[\widehat{\beta}_p]} x_{ip}\right) = 0.$$

Next, we proceed to compute the covariance matrix of $\vec{e}$. We note that

- $\vec{e} = (I_n - H)\vec{Y} = (I_n - H)(X\beta + \vec{\varepsilon}) = (I_n - H)\vec{\varepsilon}$

- Since $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ are independent normal random variables and $\varepsilon_i \sim N(0, \sigma^2)$, the covariance matrix of $\vec{\varepsilon}$ is

$$Cov(\vec{\varepsilon}) = \begin{pmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{pmatrix} = \sigma^2 I_n$$

Then the covariance matrix of $\vec{e}$ is seen to be

$$Cov(\vec{e}) = (I_n - H)(\sigma^2 I_n)(I_n - H)^T = \sigma^2 (I_n - H).$$

Hence, we deduce from equation $(*)$ that

$$\mathbb{E}\left[\sum_{i=1}^{n} (y_i - \hat{y}_i)^2\right] = \sum_{i=1}^{n} Var(e_i) = tr\big(Cov(\vec{e})\big) \overset{\substack{tr(A \pm B) = tr(A) \pm tr(B) \\ tr(cA) = c \cdot tr(A)}}{=} \sigma^2 \big(tr(I_n) - tr(H)\big)$$

$$= \sigma^2 \big(n - tr(X(X^T X)^{-1} X^T)\big) \overset{tr(AB) = tr(BA)}{=} \sigma^2 \left(n - tr\left(\underbrace{X^T X (X^T X)^{-1}}_{I_{p+1}}\right)\right)$$

$$= \sigma^2 (n - p - 1).$$

Here, $tr(A)$ denotes the *trace* of a square matrix A which is defined as the sum of all diagonal entries (i.e. $tr(A) = a_{11} + a_{22} + \cdots + a_{nn}$).

Therefore, we deduce that

$$\mathbb{E}[s^2] = \mathbb{E}\left[\frac{1}{n-p-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2\right] = \frac{1}{n-p-1}(n-p-1)\sigma^2 = \sigma^2$$

and s^2 is the unbiased estimator for σ^2.

Proof of Property 3

To do so, we consider the following two quantities:

$$\vec{e} = \vec{Y} - \hat{Y}, \qquad \vec{f} = X\hat{\beta} - X\vec{\beta}.$$

Since s^2 can be expressed as $s^2 = \frac{1}{n-p-1}\vec{e}^T\vec{e}$, so one can prove the property by arguing that $\vec{e}$ and $\vec{f}$ are independent.

We write $\varepsilon = \sigma\vec{Z}$ where $\vec{Z} = \begin{pmatrix} Z_1 \\ \vdots \\ Z_n \end{pmatrix}$ is a vector of independent standard normal random variables. Note that

$$\vec{e} = \vec{Y} - \hat{Y} = (I_n - H)\vec{\varepsilon} = \sigma(I_n - H)\vec{Z};$$

$$\vec{f} = X\hat{\beta} - X\vec{\beta} = H\vec{Y} - X\vec{\beta} = HX\vec{\beta} + H\vec{\varepsilon} - X\vec{\beta} \overset{H=X(X^TX)^{-1}X^T}{=\!=} X\vec{\beta} + \sigma H\vec{Z} - X\vec{\beta} = \sigma H\vec{Z}.$$

It follows from the definition that

$$\overrightarrow{W} = \begin{pmatrix} \vec{e} \\ \vec{f} \end{pmatrix} = \underbrace{\begin{pmatrix} \sigma(I_n - H) \\ \sigma H \end{pmatrix}}_{B} \vec{Z}$$

has multivariate normal distribution.

According to the property 5 of multivariate normal distribution, it is sufficient to show that the covariance matrix $\Sigma_{ef} = cov(\vec{e}, \vec{f})$ is 0. By direct computation, we get

$$\Sigma_{ef} = \mathbb{E}\left[\begin{pmatrix} e_1 - \mathbb{E}[e_1] \\ e_2 - \mathbb{E}[e_2] \\ \vdots \\ e_n - \mathbb{E}[e_n] \end{pmatrix} (f_1 - \mathbb{E}[f_1] \quad f_2 - \mathbb{E}[f_2] \quad \cdots \quad f_n - \mathbb{E}[f_n])\right]$$

$$\overset{\mathbb{E}[e_i]=\mathbb{E}[f_i]=0}{\triangleq} \mathbb{E}\left[\vec{e}\vec{f}^T\right] = \sigma^2(I_n - H)\underbrace{\mathbb{E}[ZZ^T]}_{=^{(*)}I_n} H = \sigma^2(H - H^2) \overset{H=H^2}{\triangleq} 0.$$

*Note: $\mathbb{E}[ZZ^T] = \begin{pmatrix} \mathbb{E}[Z_1^2] & \mathbb{E}[Z_1Z_2] & \cdots & \mathbb{E}[Z_1Z_n] \\ \mathbb{E}[Z_2Z_1] & \mathbb{E}[Z_2^2] & \cdots & \mathbb{E}[Z_2Z_n] \\ \vdots & \vdots & \ddots & \vdots \\ \mathbb{E}[Z_nZ_1] & \mathbb{E}[Z_nZ_2] & \cdots & \mathbb{E}[Z_n^2] \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$.

The proof is completed.

Proof of Property 4

Note that

- The matrix H and $I_n - H$ are symmetric since
$$H^T = (X(X^TX)^{-1}X^T)^T = X((X^TX)^T)^{-1}X^T = X(X^TX)^{-1}X^T = H$$
$$(I_n - H)^T = I_n - H^T = I_n - H.$$
- $H^2 = H$ so that $(I_n - H)^2 = (I_n - H)$.
- We let λ be the eigenvalue of $I_n - H$ and $\vec{v} \neq 0$ be the corresponding eigenvector, then we have
$$(I_n - H)\vec{v} = \lambda\vec{v} \Rightarrow \underbrace{(I_n - H)^2}_{=(I_n - H)}\vec{v} = \lambda\underbrace{(I_n - H)\vec{v}}_{\lambda\vec{v}} \Rightarrow \lambda\vec{v} = \lambda^2\vec{v} \Rightarrow \lambda(\lambda - 1)\vec{v} = \vec{0}$$
$$\Rightarrow \lambda = 0 \ \ or \ \ 1.$$
So the eigenvalue of $I_n - H$ can only be 0 or 1.

Since $I_n - H$ is also symmetric, then the matrix $I_n - H$ can be diagonalized into

$$I_n - H = Q\underbrace{\begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}}_{D}Q^T,$$

where $Q = (\vec{v}_1 \quad \cdots \quad \vec{v}_n)$ is the matrix which the columns contain all eigenvectors of $I_n - H$. As $I_n - H$ is symmetric, so Q is orthogonal and $QQ^T = Q^TQ = I_n$.

Recall that $tr(I_n - H) = n - p - 1$ (from the proof of property 2), it follows that

$$tr(QDQ^T) = n - p - 1 \Rightarrow tr\left(\underbrace{Q^T Q}_{=I_n} D\right) = n - p - 1 \Rightarrow \lambda_1 + \cdots + \lambda_n = n - p - 1.$$

Since $\lambda_i = 0$ or 1, this implies that $\lambda_1 = \lambda_2 = \cdots = \lambda_{n-p-1} = 1$ and $\lambda_{n-p} = \cdots = \lambda_n = 0$.

On the other hand, we consider

$$Q^T \vec{e} = \sigma Q^T \underbrace{(QDQ^T)}_{I_n - H} \vec{Z} = \sigma D Q^T \vec{Z}.$$

Since $\vec{Z} = \begin{pmatrix} Z_1 \\ \vdots \\ Z_n \end{pmatrix}$ is vector of independent standard normal random variable and $cov\left(Q^T \vec{Z}\right) =$

$Q^T I_n Q = I_n$. It follows from the properties of multivariate normal distribution that $Q^T \vec{Z} = \begin{pmatrix} Z_1' \\ \vdots \\ Z_n' \end{pmatrix} =$

$\vec{Z}'$ is also a vector of independent standard normal random variables (why?).

Therefore, we deduce that

$$\frac{(y_i - \hat{y}_i)^2}{\sigma^2} = \frac{\vec{e}^T \vec{e}}{\sigma^2} = \frac{(Q^T \vec{e})^T (Q^T \vec{e})}{\sigma^2} = \left(D\vec{Z}'\right)^T \left(D\vec{Z}'\right) = (Z_1')^2 + (Z_2')^2 + \cdots + \left(Z_{n-p-1}'\right)^2.$$

So $\frac{(y_i - \hat{y}_i)^2}{\sigma^2} \sim \chi^2_{n-p-1}$ by definition.

Example 13 (Confidence interval of $\widehat{\beta}_i$)

Suppose that the return of a stock (0005.HK) is governed by

$$r - r_f = \beta_0 + \beta_1(r_M - r_f) + \varepsilon,$$

where r_f is the riskfree interest rate, r_M is the return of market portfolio (which is a portfolio consists of all assets traded in the market) and $\varepsilon \sim N(0, \sigma^2)$ is the random error. Assuming that $r_f = \frac{0.02}{252}$.

To estimate the coefficient β, the historical returns of the asset and market portfolio (Hangseng Index) in the past 7 years are taken and analysed.

(a) Find the least-square estimate of β_1 (i.e. $\hat{\beta}_1$)

(b) Hence, find the 95% confidence interval of β_1.

😊Solution

(a) We let $y_i = r_i - r_f$ and $x_i = r_M - r_f$. Then the formula becomes

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon.$$

Given the sample data on r_i, r_f and r_M, we calculate the sample values of excess returns $r_i - r_f$ and $r_M - r_f$. (*Here, the sample size is 1723)

Using the regression formula, $\hat{\beta}$ is found to be

$$\hat{\beta} \approx \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} -0.0002239 \\ 0.7694127 \end{pmatrix}$$

```
Call:
lm(formula = y ~ x)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0002239  0.0001797  -1.246    0.213
x            0.7694127  0.0169954  45.272   <2e-16 ***
```

(b) To compute the confidence interval for β_1, we note that

- By property 1, we note that $\hat{\beta}_1$ is normally distributed with mean β_1 and variance $\sigma^2 c_{11}$, where c_{jj} denotes $(j+1)^{th}$ diagonal entries of $(X^T X)^{-1}$.
- Since σ is unknown, we have to replace σ by s. Note that s and $\hat{\beta}$ are independent, one can deduce that

$$T = \frac{\hat{\beta}_1 - \beta_1}{s\sqrt{c_{jj}}} = \frac{\frac{\hat{\beta}_1 - \beta_1}{\sigma\sqrt{c_{jj}}}}{\sqrt{\frac{s^2}{\sigma^2}}} = \frac{\frac{\hat{\beta}_1 - \beta_1}{\sigma\sqrt{c_{jj}}}}{\sqrt{\frac{s^2}{\sigma^2}}} = \frac{\frac{\hat{\beta}_1 - \beta_1}{\sigma\sqrt{c_{jj}}}}{\sqrt{\frac{1}{n-p-1}\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sigma^2}}} \sim t_{n-p-1}.$$

So $T = \frac{\hat{\beta}_1 - \beta_1}{s\sqrt{c_{jj}}}$ has T-distribution with degree of freedom $n - p - 1$.

Next, we find the critical values t_L and t_U such that

$$P(t_L \leq T \leq t_U) = 0.95$$

Take $P(T \leq t_L) = P(T \geq t_U) = 0.025$ and $n - p - 1 = 1721$, we get

$$t_L = -1.961343, \qquad t_U = 1.961343.$$

Thus, we deduce that

$$P\left(-1.961343 \leq \frac{\hat{\beta}_1 - \beta_1}{\underbrace{s\sqrt{c_{jj}}}_{=0.0169954}} \leq 1.961343\right) = 0.95$$

$$\Rightarrow P(0.7360789 \leq \beta_1 \leq 0.8027465) = 0.95.$$

So the 95% confidence interval is $[0.7360789,\ 0.8027465]$.

Example 14 (Confidence interval of mean response of y_t)

We consider the linear regression model $y_t = \beta_0 + \beta_1 x_{t1} + \cdots + \beta_p x_{tp} + \varepsilon$.

We let $\left(y_i, x_{i1}, x_{i2}, \ldots, x_{ip}\right)_{1 \le i \le n}$ be the sample data.

Given the input variable $\vec{x}^T = \left(1, x_1, x_2, \ldots, x_p\right)$, find the $(100\alpha)\%$ confidence interval for the *mean response*, i.e.

$$\mathbb{E}[y_t] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p = \vec{x}^T \vec{\beta}.$$

😊Solution

Recall that $\hat{\beta} \sim N_{p+1}\left(\vec{\beta}, \sigma^2 (X^T X)^{-1}\right)$, it follows from the property of multivariate normal distribution that

$$\vec{x}^T \hat{\beta} \sim N\left(\vec{x}^T \vec{\beta},\ \ \sigma^2 \vec{x}^T (X^T X)^{-1} \vec{x}\right).$$

$$\Rightarrow Z = \frac{\vec{x}^T \hat{\beta} - \vec{x}^T \vec{\beta}}{\sqrt{\sigma^2 \vec{x}^T (X^T X)^{-1} \vec{x}}} \sim N(0,1).$$

As σ is unknown, one shall replace σ by the sample variance s. Since s and $\hat{\beta}$ are independent, one can show that

$$T = \frac{\vec{x}^T\hat{\beta} - \vec{x}^T\vec{\beta}}{\sqrt{s^2\vec{x}^T(X^TX)^{-1}\vec{x}}} = \frac{\frac{\vec{x}^T\hat{\beta} - \vec{x}^T\vec{\beta}}{\sqrt{\sigma^2\vec{x}^T(X^TX)^{-1}\vec{x}}}}{\sqrt{s^2/\sigma^2}} = \frac{Z}{\sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{(n-p-1)\sigma^2}}} = \frac{Z}{\sqrt{\frac{\chi^2_{n-p-1}}{n-p-1}}} \sim t_{n-p-1}.$$

Given this result, we proceed to find t_L, t_U such that

$$P\left(t_L \le T_{n-p-1} \le t_U\right) = \alpha.$$

Then we deduce that

$$P\left(t_L \le \frac{\vec{x}^T\hat{\beta} - \vec{x}^T\vec{\beta}}{\sqrt{s^2\vec{x}^T(X^TX)^{-1}\vec{x}}} \le t_U\right) = \alpha.$$

$$\Rightarrow P\left(\vec{x}^T\hat{\beta} - t_U\sqrt{s^2\vec{x}^T(X^TX)^{-1}\vec{x}} \le \vec{x}^T\vec{\beta} \le \vec{x}^T\hat{\beta} - t_L\sqrt{s^2\vec{x}^T(X^TX)^{-1}\vec{x}}\right) = \alpha$$

Thus the $(100\alpha)\%$ confidence interval of $\vec{x}^T\vec{\beta}$ is

$$\left[\vec{x}^T\hat{\beta} - t_U\sqrt{s^2\vec{x}^T(X^TX)^{-1}\vec{x}},\ \ \vec{x}^T\hat{\beta} - t_L\sqrt{s^2\vec{x}^T(X^TX)^{-1}\vec{x}}\right].$$

Example 15 (Prediction interval of y_t)

Given the input variable $\vec{x}^T = (1, x_1, x_2, \dots, x_p)$, find the $(100\alpha)\%$ confidence interval (known as prediction interval) of the *actual value* of y.

*Note: Recall that y satisfies $\vec{y} = \vec{x}^T\vec{\beta} + \varepsilon$. So the actual value of y is also affected by an unobservable random error ε.

☺Solution

Similar to earlier example, we first study the distribution of the estimator $\vec{x}^T\hat{\beta}$.

We note the following:

- From the previous example, we see that $\vec{x}^T\hat{\beta} \sim N\left(\vec{x}^T\vec{\beta},\ \sigma^2\vec{x}^T(X^TX)^{-1}\vec{x}\right)$.
- Since $\varepsilon \sim N(0, \sigma^2)$, then $y \sim N\left(\vec{x}^T\vec{\beta},\ \sigma^2\right)$.
- It follows that

$$y - \vec{x}^T\hat{\beta} \sim N\left(0, \sigma^2(1 + \vec{x}^T(X^TX)^{-1}\vec{x})\right)$$

$$\Leftrightarrow \frac{y - \vec{x}^T\hat{\beta}}{\sqrt{\sigma^2(1 + \vec{x}^T(X^TX)^{-1}\vec{x})}} \sim N(0,1).$$

Replacing σ by sample SD s, one can show that

$$\frac{y-\vec{x}^T\hat{\beta}}{\sqrt{s^2(1+\vec{x}^T(X^TX)^{-1}\vec{x})}}\sim t_{n-p-1}.$$

Given t_L, t_U that satisfies $P\left(t_L \le t_{n-p-1} \le t_U\right) = \alpha$. Then we have

$$P\left(t_L \le \frac{y-\vec{x}^T\hat{\beta}}{\sqrt{s^2(1+\vec{x}^T(X^TX)^{-1}\vec{x})}} \le t_U\right) = \alpha$$

$$\Rightarrow P\left(\vec{x}^T\hat{\beta} + t_L\sqrt{s^2(1+\vec{x}^T(X^TX)^{-1}\vec{x})} \le y \right.$$
$$\left. \le \vec{x}^T\hat{\beta} + t_U\sqrt{s^2(1+\vec{x}^T(X^TX)^{-1}\vec{x})}\right) = \alpha.$$

The $(100\alpha)\%$ confidence interval is

$$\left[\vec{x}^T\hat{\beta} + t_L\sqrt{s^2(1+\vec{x}^T(X^TX)^{-1}\vec{x})}, \vec{x}^T\hat{\beta} + t_U\sqrt{s^2(1+\vec{x}^T(X^TX)^{-1}\vec{x})}\right].$$

Comparing with the confidence interval of $\mathbb{E}[y]$ in earlier example, we observe that the confidence interval of y is wider. It is because the value of y is also affected by random error ε which increases the uncertainty on y.